

Rich-Neighborhood Contrastive Learning Framework for Drug Repositioning via Structural and Semantic Neighbor Fusion

Appendix

Fig. S1. Brimonidine-LRRK2 (2ZEJ) docking: 3D binding pose (left) and 2D interaction map (right).

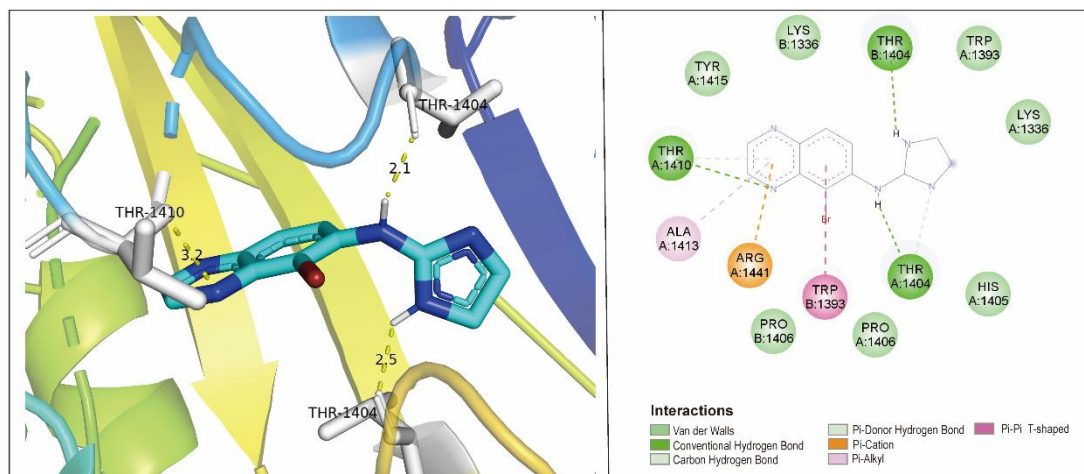


Fig. S2. Pamidronic acid-GRM5 (5CGD) docking: 3D binding pose (left) and 2D interaction map (right).

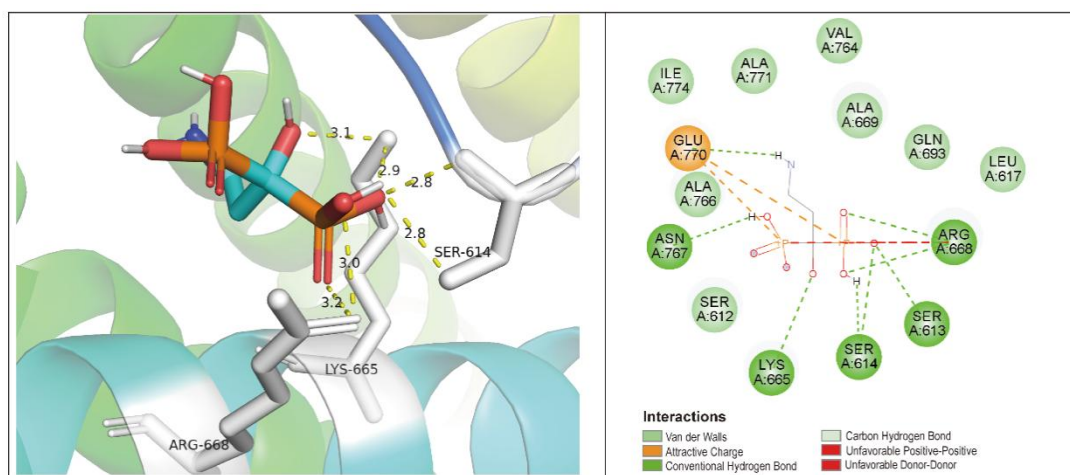
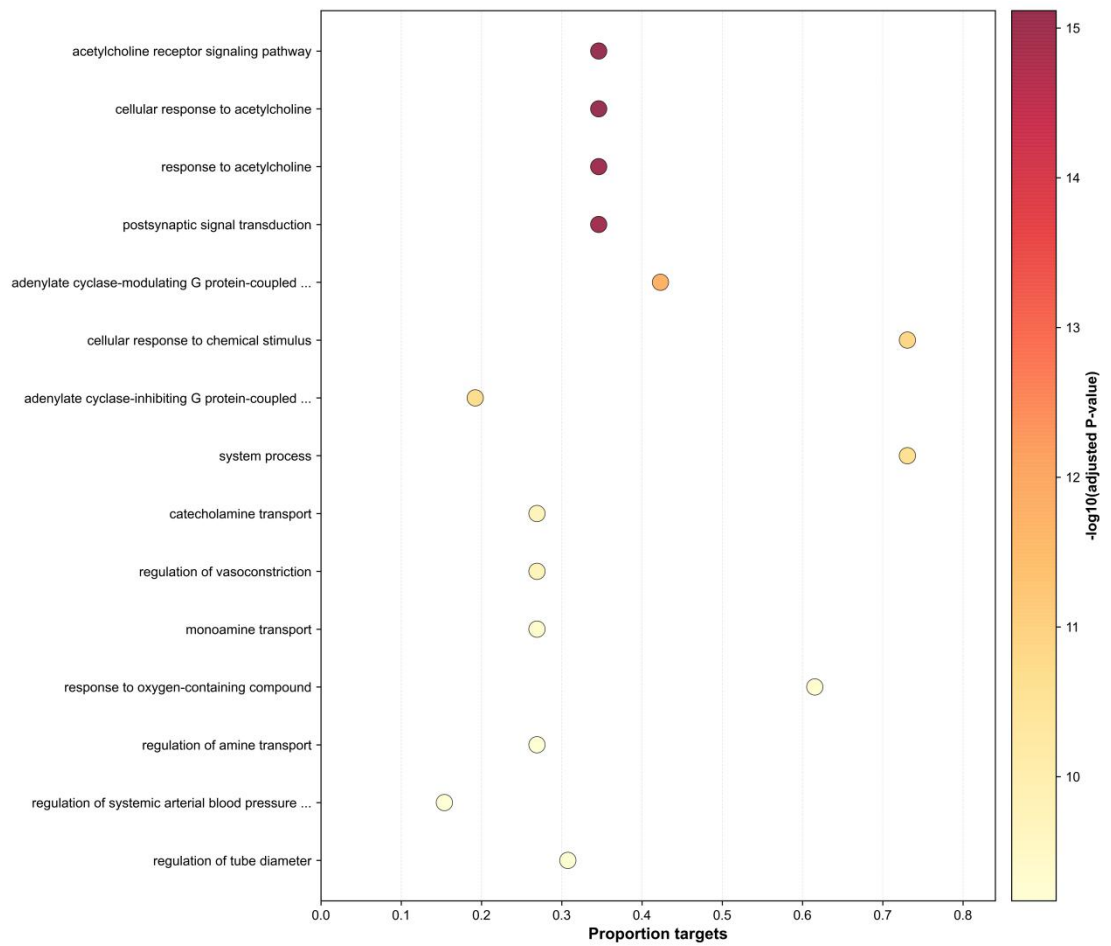


Fig S3. Enriched gene ontology terms (Biological Process) among all predicted PD drug targets. The x axis shows the proportion of targets mapped to each pathway.



Supplementary Table S1: Representative Drugs from Three Largest Clusters

Cluster	Size	Prototype Class	Purity	Representative Drugs
2	52	Antineoplastic (ATC: L)	90.4%	Bortezomib, Imatinib, Methotrexate
37	33	Nervous System (ATC: N)	90.9%	Morphine, Fentanyl, Tramadol
18	26	Cardiovascular (ATC: C)	96.2%	Valsartan, Losartan, Enalapril
Average purity:92.5%				

Supplementary Table S2: Computational resources and efficiency on experimental datasets

Dataset	Nodes	Edges	Edges(s)	Time/Epoch(s)	Peak GPU(GB)
Fdataset	909	1933	0.35	0.48	1.23
Cdataset	1072	2532	0.36	0.50	1.28
Ldataset	865	18390	0.36	1.11	1.43